

CLM/Modbus Parameter Reference Dictionary

LK09-240

Communication
Link Module

***CLM/Modbus
Parameter Reference
Dictionary***

LK09-240
Release (320/400)
8/93

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About This Publication

This publication is a reference for process engineers, control system engineers, and application engineers who design and implement the Honeywell Communication Link Module on an Application Module of the TDC 3000^X, Network. This publication provides reference information about parameters, a dictionary of filenames and explanations, and summary information on CL programs and data segments.

This publication is part of a set of publications that provide parameter reference information for nodes on the LCN. The other members of this set are:

- *Application Module Parameter Reference Dictionary* in the *Implementation/AM* - 1 binder
- *Process Manager Parameter Reference Dictionary* in the *Implementation/Process Manager* - 2 binder
- *Logic Manager Parameter Reference Dictionary* in the *Implementation/LM* - 2 binder
- *Hiway Gateway Parameter Reference Dictionary* in the *Implementation/Hiway Gateway* - 1 binder
- *Programmable Logic Controller Gateway Parameter Reference Dictionary* in the *Implementation/PLC Gateway* binder
- *Computer Gateway Parameter Reference Dictionary* in the *Implementation/Computer Gateway* binder

This publication supports TDC 3000^X, software release 320.

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References

Configuring the AM

Publication Title	Publication Number	Binder Title	Binder Number
<i>Application ModuleForms</i>	AM88-300	Implementation/Configuration Forms	TDC 1031
<i>Application Module Form Instructions</i>	AM12-300	Implementation/AM - 1	TDC 1035-1
<i>Application Module Control Functions</i>	AM09-302	Implementation/AM - 1	TDC 1035-1
<i>Application Module Algorithm Engineering Data</i>	AM09-301	Implementation/AM - 1	TDC 1035-1
<i>Engineer's Reference Manual</i>	SW09-305	Implementation/Startup & Reconfiguration - 2	TDC 1030-2
<i>Data Entity Builder Manual</i>	SW11-311	Implementation/Engineering Operations - 1	TDC 1032

Programming in CL

Publication Title	Publication Number	Binder Title	Binder Number
<i>Control Language/Application Module Overview</i>	SW27-300	Implementation/AM - 2	TDC 1035-2
<i>Control Language/Application Module Reference Manual</i>	AM27-310	Implementation/AM - 2	TDC 1035-2
<i>Control Language/Application Module Data Entry</i>	AM11-385	Implementation/AM - 2	TDC 1035-2

Section 1 – Introduction

1.1 General

This section

This section covers the following topics.

Topic		See Page
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1.1	General.....	1
1.2	Use of This Publication.....	1

What this manual is

This publication contains reference information on the TDC 3000^X, System parameters that are associated with Honeywell's Communication Link Module (CLM).

The reference information includes three types of data:

1. a dictionary that includes all the parameter names associated with the CLM; for each parameter name listed, related data is given (such as a description, default value, and permissible range of values)
 2. a dictionary of file names supplied in the release
 3. a summary of the names and descriptions of the CL programs and Custom Data Segments in the CLM package.
-

1.2 Use of This Publication

When to use it

This publication should be used for reference when

- building AM data points
 - planning and writing CL programs associated with the AM
 - parameter names appear on the Universal Station displays when the system is in operation.
-

Section 2 – Parameter Dictionary

2.1 General

This section

This section consists of a dictionary that includes all the parameter names associated with the CLM; for each parameter name listed, related data is given (such as a description, default value, and permissible range of values).

This section covers the following topics.

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2.2 Parameter

Definition

A parameter name is a capitalized character string of one to eight alphabetic or alphanumeric characters.

A parameter is a mnemonic which identifies a value used in the system.

2.3 Format

Overview

The standard form of each parameter entry is presented in the format example below.

Source is the name of the CDS module or AM in which the parameter is defined.

Entry format

Point Type AM Regulatory

Source **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

2.4 Parameter Name Listing

ACTION

Point Type AM Custom, Builder Point (BUILD_nn)

Builder Action Code

Source BUILD_CDS.CL **Default** 0 **Access Lock** Engineer

Type Real

Range >= 0.0

ALMOPT

Point Type AM Switch

Discrete Alarm Option

Source DIG2_CDS.CL **Default** 0 **Access Lock** Engineer
DIG3_CDS.CL build visible
DIG4_CDS.CL
DIG5_CDS.CL

Type Real

Range 0=none
1=offnorml
2=cmddis

AV

Point Type AM Regulatory

Totalizer Point Input value

Source M_TZ.CL **Default** 0 **Access Lock** Engineer

Type Real

Range Any real number

BREAK

Point Type AM Custom, Builder Point (BUILD_nn)

Interrupt File Input/BackBuild

Source BUILD_CDS.CL **Default** OFF **Access Lock** Engineer

Type Logical

Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

BUILDER	Point Type AM Custom, Builder Point (BUILD_nn) Builder display invoked Source BUILD_CDS.CL Default OFF Access Lock Engineer Type Logical Range OFF, ON
C1 C2	Point Type AM Regulatory Totalizer Point conversion information Source M_TZ.CL Default 0 Access Lock Engineer Type Real Range >= 0.0
CKPTCYCL	Point Type AM Custom, Network Point (CLM_nn) Checkpoint cycle period Source NET_CDS.CL Default 4.0 Access Lock Engineer Type Real Range >= 0.0
CKPTSTRT	Point Type AM Custom, Network Point (CLM_nn) Checkpoint start offset Source NET_CDS.CL Default 30.0 Access Lock Engineer Type Real Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

CLEARALM

Point Type AM Custom, Network Point (CLM_nn)

CLM Alarm clear indicator

Source NETCDS.CL **Default** OFF **Access Lock** Engineer

Type Logical

Range OFF, ON

CLMFAIL

Point Type AM Custom, Network Point (CLM_nn)

Composite CLM failure indicator

Source NETCDS.CL **Default** 0 **Access Lock** Engineer

Type REAL

Range >= 0

CLMSTAT

Point Type AM Custom, Network Point (CLM_nn)

Composite CLM Status

Source NETCDS.CL **Default** 4 **Access Lock** Engineer

Type Real

Range 0 to 4 (GOOD/ERROR/FAILED/UNKNOWN)

CLM_DEV

Point Type AM Regulatory or AM Switch

NDT Device Name saved by the Builder

Source CLM_DMD.CL **Default** "" **Access Lock** Engineer
CLM_DMD2.CL (build visible)

Type String

Continued on next page

2.4 Parameter Name Listing, Continued

CLM_OUT	Point Type AM Regulatory or AM Switch
	Storing to CLM_OUT causes PITE outputs to process
Source	CLM_DMD.CL Default OFF Access Lock Engineer CLM_DMD2.CL
Type	Logical
Range	OFF, ON

CLM_PIT	Point Type AM Regulatory or AM Switch
	NDT PIT table Name saved by the Builder
Source	CLM_DMD.CL Default "" Access Lock Engineer CLM_DMD2.CL (build visible)
Type	String

CLM_PORT	Point Type AM Regulatory or AM Switch
	NDT Port Device Name saved by the Builder
Source	CLM_DMD.CL Default "" Access Lock Engineer CLM_DMD2.CL (build visible)
Type	String

CLM_PPS	Point Type AM Regulatory or AM Switch
	Storing to CLM_PPS causes PITE to process
Source	CLM_DMD.CL Default OFF Access Lock Engineer CLM_DMD2.CL
Type	Logical
Range	OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

COLFLG()	Point Type AM Regulatory
	Buffer Collection Flag
Source	NETCDS.CL Default 0.0 Access Lock Operator
Type	Real Array (1..4)
Range	>= 0.0

CPPTCLM	Point Type AM Custom, Network Point (CLM_nn)
	Checkpoint this CLM
Source	NETCDS.CL Default OFF Access Lock Operator
Type	Logical
Range	OFF, ON

CV	Point Type AM Regulatory
	The result of control algorithm processing prior to output processing
Source	AM Default 0 Access Lock Program
Type	Real
Range	Any real number

Continued on next page

2.4 Parameter Name Listing, Continued

D1

Point Type AM Switch

Input 1 status

Source DIG2CDS.CL **Default** 0 **Access Lock** Engineer
DIG3CDS.CL
DIG4CDS.CL
DIG5CDS.CL
DIN2CDS.CL
DIN4CDS.CL

Type Logical

Range OFF, ON

D1_0

Point Type AM Switch

D1_Zero PV State (NODINPTS=1)

Source DIG3CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG4CDS.CL build visible
DIG5CDS.CL

Type Self-Defining Enumeration

Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4

Note: For 0 input conn., \$ord0=S1STATES(0) describes D1=0
 \$ord1=S1STATES(1) describes D1=0
 \$ord2=S1STATES(2) describes D1=0
 \$ord3=S1STATES(3) describes D1=0
 \$ord4=S1STATES(4) describes D1=0
 otherwise

Note: For 4-state points, \$ord3 must be the user's description of bad PV, \$ord2 is usually for moving.
For 5-state points, \$ord4 must be the user's description of bad PV, \$ord3 is usually for moving.

Continued on next page

2.4 Parameter Name Listing, Continued

D1_1

Point Type AM Switch

D1_One PV State (NODINPTS=1)

Source DIG2CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG3CDS.CL
DIG4CDS.CL
DIG5CDS.CL

Type Self-Defining Enumeration

Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4

Note: For 0 input conn., \$ord0=S1STATES(0) describes D1=1

\$ord1=S1STATES(1) describes D1=1

\$ord2=S1STATES(2) describes D1=1

\$ord3=S1STATES(3) describes D1=1

otherwise \$ord4=S1STATES(4) describes D1=1

Note: For 4-state points, \$ord3 must be the user's description of bad PV, \$ord2 is usually for moving.

For 5-state points, \$ord4 must be the user's description of bad PV, \$ord3 is usually for moving.

D2

Point Type AM Switch

Input 2 status

Source DIG4CDS.CL **Default** OFF **Access Lock** Engineer
DIG5CDS.CL
DIN4CDS.CL

Type Logical

Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

D2D1_00

Point Type AM Switch

D2_D1 Zero_Zero PV State (NODINPTS=2)

Source DIG2CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG3CDS.CL build visible
DIG4CDS.CL
DIG5CDS.CL

Type Self-Defining Enumeration

Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4

Note: For 4-state points, \$ord3 must be the user's description of bad PV, \$ord2 is usually for moving.
For 5-state points, \$ord4 must be the user's description of bad PV, \$ord3 is usually for moving.

D2D1_01

Point Type AM Switch

D2_D1 Zero_One PV State (NODINPTS=2)

Source DIG2CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG3CDS.CL build visible
DIG4CDS.CL
DIG5CDS.CL

Type Self-Defining Enumeration

Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4

Note: For 4-state points, \$ord3 must be the user's description of bad PV, \$ord2 is usually for moving.
For 5-state points, \$ord4 must be the user's description of bad PV, \$ord3 is usually for moving.

Continued on next page

2.4 Parameter Name Listing, Continued

D2D1_10

Point Type AM Switch

D2_D1 One_Zero PV State (NODINPTS=2)

Source DIG2CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG3CDS.CL build visible
DIG4CDS.CL
DIG5CDS.CL

Type Self-Defining Enumeration

Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4

Note: For 4-state points, \$ord3 must be the user's description of bad PV, \$ord2 is usually for moving.
For 5-state points, \$ord4 must be the user's description of bad PV, \$ord3 is usually for moving.

D2D1_11

Point Type AM Switch

D2_D1 One_One PV State (NODINPTS=2)

Source DIG2CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG3CDS.CL build visible
DIG4CDS.CL
DIG5CDS.CL

Type Self-Defining Enumeration

Range \$ord0, \$ord1, \$ord2, \$ord3, \$ord4

Note: For 4-state points, \$ord3 must be the user's description of bad PV, \$ord2 is usually for moving.
For 5-state points, \$ord4 must be the user's description of bad PV, \$ord3 is usually for moving.

Continued on next page

2.4 Parameter Name Listing, Continued

DATABUF()	Point Type AM Custom, Builder Point (BUILD_nn)
	Miscellaneous Data Usage array
Source	BLD2CDS.CL Default 0 Access Lock Engineer
Type	Real ARRAY(1..20)
Range	>= 0.0

DEBUGLEV	Point Type AM Custom, Builder and Network Points (BUILD_nn and CLM_nn)
	Debug Level for program tracing. Messages are displayed based on this value > 5 (Process tracing and status check)
Source	BUILDCDS.CL Default 0 Access Lock Engineer NETCDS.CL
Type	Real
Range	>= 0.0

DEFARR()	Point Type AM Custom, Builder Point (BUILD_nn)
	Definition Data Array
Source	BUILDCDS.CL Default "" Access Lock Engineer
Type	ARRAY(1..50) of String

Continued on next page

2.4 Parameter Name Listing, Continued

DEFNUM

Point Type AM Custom, Builder Point (BUILD_nn)

Number of definitions this entry

Source BUILDCDS.CL **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

**DEMNDIN1,
DEMNDIN2,
DEMNDIN3,
DEMNDIN4**

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) demand address index #

Source NETCDS1.CL (DEMNDIN1) **Default** 0 **Access Lock** Oper
NETCDS2.CL (DEMNDIN2)
NETCDS3.CL (DEMNDIN3)
NETCDS4.CL (DEMNDIN4)

Type Real

Range ≥ 0.0

DETSTAT

Point Type AM Custom, Builder Point (BUILD_nn)

Error Detailed Status

Source BUILDCDS.CL **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

ENTARR()	Point Type AM Custom, Builder Point (BUILD_nn) Entry Type Array Source BLD2CDS.CL Default 0 Access Lock Engineer Type ARRAY(1..120) of String
ENTRY	Point Type AM Custom, Builder Point (BUILD_nn) Entry Source BUILD_CDS.CL Default 0 Access Lock Engineer Type Real Range 0=none,1=NDT,2=PIT,3=PITE,4=PITG
ENTRYNAM	Point Type AM Custom, Builder Point (BUILD_nn) Entry Name Source BUILD_CDS.CL Default spaces Access Lock Engineer Type String
ENTRYTYP	Point Type AM Custom, Builder Point (BUILD_nn) Entry Type (TDF NAME) Source BUILD_CDS.CL Default spaces Access Lock Engineer Type String

Continued on next page

2.4 Parameter Name Listing, Continued

**FB1,
FB2,
FB3**

Point Type AM Switch

Output 1 (2,3) Signals as read back from the RTU reset terminals.

Source DIG2CDS.CL **Default** OFF **Access Lock** Program
DIG3CDS.CL
DIG4CDS.CL
DIG5CDS.CL

Type Logical

Range OFF, On

FBTIME

Point Type AM Switch

Feedback time

Source DIG2CDS.CL **Default** NAN **Access Lock** Engineer
DIG3CDS.CL build visible
DIG4CDS.CL
DIG5CDS.CL

Type Real

Range >= 0.0

FMTARR()

Point Type AM Custom, Builder Point (BUILD_nn)

Format Data Array

Source BUILD_CDS.CL **Default** "" **Access Lock** Engineer

Type ARRAY(1..50) of String

Continued on next page

2.4 Parameter Name Listing, Continued

GENATTR()

Point Type AM Custom

Display attributes

Source GENRLCDS.CL **Default** 6 **Access Lock** Engineer

Type ARRAY (1..40) of NUMBER

Range 1 to 8

CODE	COLOR	BLINK	REVERSE	COLOR & TARGET
1	RED	11	21	101
2	YELLOW	12	22	102
3	GREEN	13	23	103
4	MAGENTA	14	24	104
5	BLUE	15	25	105
6	CYAN	16	26	106
7	WHITE	17	27	107
8	BLACK	NO	NO	108

GENCOM

Point Type AM Custom

Display command

Source GENRLCDS.CL **Default** 0 **Access Lock** Engineer

Type NUMBER

Range >= 0

GENDISP()

Point Type AM Custom

Display strings

Source GENRLCDS.CL **Default** spaces **Access Lock** Engineer

Type NUMBER

Range ARRAY (1–40) of String

GENNODE

Point Type AM Custom

Display node identification

Source GENRLCDS.CL **Default** **Access Lock** Engineer

Type NUMBER

Range >= 0

Continued on next page

2.4 Parameter Name Listing, Continued

GENSTAT

Point Type AM Custom

Display status

Source GENRLCDS.CL **Default** 0 **Access Lock** Engineer

Type NUMBER

Range >= 0

GENSTATE

Point Type AM Custom

Display state

Source GENRLCDS.CL **Default** 0 **Access Lock** Engineer

Type NUMBER

Range >= 0

GENSUB

Point Type AM Custom

Display subpicture used

Source GENRLCDS.CL **Default** 0 **Access Lock** Engineer

Type NUMBER

Range 1 to 3

1 = 16 character strings/120 per display/40 from each LCN point

2 = 20 character strings/96 per display/32 from each LCN point

3 = 40 character strings/48 per display/16 from each LCN point

GENTAR1

Point Type AM Custom

Target string 1

Source GENRLCDS.CL **Default** " APPL " **Access Lock** Engineer

Type STRING

Continued on next page

2.4 Parameter Name Listing, Continued

GENTARZ

Point Type AM Custom

Target string 2

Source GENRLCDS.CL **Default** "SCHEMATIC" **Access Lock** Engr

Type STRING

GROUPIN1,
GROUPIN2,
GROUPIN3,
GROUPIN4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) group address index #

Source NETCDS1.CL (GROUPIN1) **Default** 0 **Access Lock** Operator
NETCDS2.CL (GROUPIN2)
NETCDS3.CL (GROUPIN3)
NETCDS4.CL (GROUPIN4)

Type Real

Range >= 0.0

INDXNO()

Point Type AM Custom, Network Point (CLM_nn)

Buffer index number

Source NETCDS.CL **Default** 0 **Access Lock** Operator

Type REAL ARRAY (1..4)

Range >= 0.0

INLINE1,
INLINE2,
INLINE3,
INLINE4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) Entries

Source NETCDS1.CL (INLINE1) **Default** 0 **Access Lock** Operator
NETCDS2.CL (INLINE2)
NETCDS3.CL (INLINE3)
NETCDS4.CL (INLINE4)

Type Real

Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

INPFILE	Point Type AM Custom, Builder Point (BUILD_nn) Input File Pathname (User Linkage File) Source BUILDCDS.CL Default spaces Access Lock Engineer Type String
INPUT	Point Type AM Custom Input String Source M_MSG.CL Default none Access Lock Engineer Type String
INPUT	Point Type AM Regulatory Measured variable Source M_AIN.CL Default 0 Access Lock Engineer Type Real Range >= 0.0
IOARRAY(.)	Point Type AM Custom Message point I/O strings Source M_MSG.CL Default none Access Lock Engineer Type String Array (1..20)

Continued on next page

2.4 Parameter Name Listing, Continued

IOTIME(.)

Point Type AM Custom

Message point I/O times

Source M_MSG.CL **Default** 0 **Access Lock** Engineer

Type Time Array (1..20)

JOURNAL

Point Type AM Custom

Message point Operator Journal option

Source M_MSG.CL **Default** OFF **Access Lock** Engineer

Type Logical

Range OFF, ON

LASTAV

Point Type AM Regulatory

AV value from previous scan

Source M_TZ.CL **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0

LASTTIME

Point Type AM Regulatory

Time value at previous scan

Source M_TZ.CL **Default** 0 **Access Lock** Engineer

Type Real

Range ≥ 0

Continued on next page

2.4 Parameter Name Listing, Continued

LAST_IN

Point Type AM Custom

Last Input String

Source M_MSG.CL **Default** spaces **Access Lock** Engineer

Type String

LAST_OUT

Point Type AM Custom

Last Output String

Source M_MSG.CL **Default** spaces **Access Lock** Engineer

Type String

LNCTR()

Point Type AM Custom, Network Point (CLM_nn)

Work Area Number

Source NETCDS.CL

Default (1.0,2.0,3.0,4.0,5.0,6.0,7.0,8.0,9.0,10.0,11.0,12.0,13.0,14.0,15.0)

Access Lock Engineer

Type Real ARRAY (1..15)

Range 1 – 15

LOCK

Point Type AM Custom, Builder Point (BUILD_nn)

Lock Display while processing

Source BUILD_CDS.CL **Default** OFF **Access Lock** Engineer

Type Logical

Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

MODATTR	Point Type AM Regulatory Mode attribute for control algorithms Source AM Default OPERATOR Access Lock Operator Type MODATTR enumeration Range NONE, OPERATOR, PROGRAM
MODE	Point Type AM Regulatory Current mode for a control algorithm Source AM Default 0 Access Lock Engineer Type MODE Enumeration Range MAN, AUTO, CAS, NORMAL
NBUFFER()	Point Type AM Custom, Network Point (CLM_nn) NDT buffer flag Source NETCDS.CL Default 0 Access Lock Operator Type Real ARRAY (1..4) Range >= 0.0
NEWSTATE()	Point Type AM Custom, Network Point (CLM_nn) NDT State change flag Source NETCDS.CL Default OFF Access Lock Operator Type LOGICAL ARRAY (1..18) Range OFF, ON
NFAIL()	Point Type AM Custom, Network Point (CLM_nn) NDT failure indicator Source NETCDS.CL Default 0 Access Lock Engineer Type Real ARRAY (1..18) Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

NMODE	Point Type AM Regulatory Normal mode for this data point. Source AM Default None Access Lock Supervisor Type MODE enumeration Range >= 0.0
NNAME()	Point Type AM Custom, Network Point (CLM_nn) NDT PORT etc. name Source NETCDS.CL Default spaces Access Lock Engineer Type ARRAY (1..18) of String
NODINPTS	Point Type AM Switch # Dig. Input Connections (0–2). Two input connections requires at least 4 states. Source DIG2CDS.CL Default 0 Access Lock Engineer DIG3CDS.CL build visible DIG4CDS.CL DIG5CDS.CL Type Real Range 0, 1, 2
NODOPTS	Point Type AM Switch # Dig. Output Connections (0–3 <=nostates) Source DIG2CDS.CL Default 0 Access Lock Engineer DIG3CDS.CL build visible DIG4CDS.CL DIG5CDS.CL Type Real Range 0, 1, 2, 3

Continued on next page

2.4 Parameter Name Listing, Continued

NSTATE()	Point Type AM Custom, Network Point (CLM_nn) NDT State Source NETCDS.CL Default 0 Access Lock Engineer Type Real ARRAY (1..18) Range >= 0.0
NSTATUS()	Point Type AM Custom, Network Point (CLM_nn) NDT Status Source NETCDS.CL Default 0 Access Lock Engineer Type Real ARRAY (1..18) Range >= 0.0
NUMBR1(), NUMBR2(), NUMBR3(), NUMBR4()	Point Type AM Custom, Network Point (CLM_nn) Buffer # (1–4) Numbers Source NETCDS1.CL (NUMBR1) Default 0 Access Lock Operator NETCDS2.CL (NUMBR2) NETCDS3.CL (NUMBR3) NETCDS4.CL (NUMBR4) Type Real ARRAY (1..300) Range >= 0.0
NUMBR1(), NUMBR2()	Point Type AM Custom, Builder Point (BUILD_nn) Number of words 1 Number of words 2 Source CLMDCDS.CL Default 0 Access Lock Operator Type Real ARRAY (1..50) Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

ON_DISP

Point Type AM Custom

Message point is currently on display at a Universal Station

Source M_MSG.CL **Default** 0 **Access Lock** Engineer

Type Real

Range >= 0.0

OP

Point Type AM Regulatory

The output of a regulatory control algorithm, in %, after control output processing.

Source AM **Default** 0 **Access Lock** Engineer

Type Real

Range >= 0.0

OUTFLAG()

Point Type AM Custom

I/O String was output. Set true to output a string.

Source M_MSG.CL **Default** OFF **Access Lock** Engineer

Type Logical Array (1..20)

Range OFF, ON

OP1,
OP2,
OP3

Point Type AM Switch

Output 1 (2,3) Signal

Source DIG2CDS.CL **Default** OFF **Access Lock** Engineer
DIG3CDS.CL
DIG4CDS.CL
DIG5CDS.CL

Type Logical

Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

OPHILM	Point Type AM Regulatory		
	Output high limit in percent		
	Source	AM Default 105.0	Access Lock Supervisor
	Type	Real	
	Range	OPLOLM TO 106.9	
OPLOLM	Point Type AM Regulatory		
	Output low limit in percent		
	Source	AM Default −5.0	Access Lock Engineer
	Type	Real	
	Range	−6.9 to OPHILM	
OUTFILE	Point Type AM Custom, Builder Point (BUILD_nn)		
	Optional Output File Pathname		
	Source	BUILDCDS.CL Default spaces	Access Lock Engineer
	Type	String	
	OUTFILOP	Point Type AM Custom, Builder Point (BUILD_nn)	
Output File Option Selected			
Source		BUILDCDS.CL Default OFF	Access Lock Engineer
Type		Logical	
Range		OFF, ON	
OUTPUT	Point Type AM Switch		
	Entered output string		
	Source	M_MSG.CL Default spaces	Access Lock Engineer
	Type	String	

Continued on next page

2.4 Parameter Name Listing, Continued

OUTPUT

Point Type AM Regulatory

Output analog value to Modbus

Source M_AO.CL **Default** 0 **Access Lock** Engineer
class general

Type Real

Range ≥ 0.0

PASCALSW

Point Type AM Custom, Network Point (CLM_nn)

Pascal Run Switch

Source NETCDS.CL **Default** ON **Access Lock** Engineer

Type Logical

Range OFF, ON

PATHIND

Point Type AM Regulatory

Control path indicator. Indicates whether control information is currently propagated normally, on hold, in initialization, or an override path is selected.

Source AM **Default** MAN **Access Lock** Read Only

Type PATHIND enumeration

Range FWD, HOLD, INIT, OR

Continued on next page

2.4 Parameter Name Listing, Continued

PITNAIN1,
PITNAIN2,
PITNAIN3,
PITNAIN4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1-4) PIT Name address index #

Source NETCDS1.CL (PITNAIN1) **Default** 0 **Access Lock** Oper
NETCDS2.CL (PITNAIN2)
NETCDS3.CL (PITNAIN3)
NETCDS4.CL (PITNAIN4)

Type Real

Range >= 0.0

PITNAM

Point Type AM Custom, Builder Point (BUILD_nn)

Assoc PIT Name for PITE

Source BUILD_CDS.CL **Default** spaces **Access Lock** Engineer

Type String

PORTCD()

Point Type AM Custom, Network Point (CLM_nn)

Buffer port code

Source NETCDS.CL **Default** 0 **Access Lock** Operator

Type Real ARRAY (1..4)

Range >= 0.0

PROCNAM

Point Type AM Custom, Builder Point (BUILD_nn) or Network Point (CLM_nn)

Program/Process Identification

Source BUILD_CDS.CL **Default** Spaces **Access Lock** Engineer
NETCDS.CL

Type String

Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

PROCOPT

Point Type AM Custom, Builder Point (BUILD_nn)

Process Option for Multiple Build

Source BUILD_CDS.CL **Default** 0 **Access Lock** Engineer

Type Real

Range >= 0.0

PTPARIN1, PTPARIN2, PTPARIN3, PTPARIN4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) point.par address index #

Source NET_CDS1.CL (PTPARIN1) **Default** 0 **Access Lock** Oper.
NET_CDS2.CL (PTPARIN2)
NET_CDS3.CL (PTPARIN3)
NET_CDS4.CL (PTPARIN4)

Type Real

Range >= 0.0

PVNORMAL

Point Type AM Switch

Contains a character string that represents the position of the normal state of the point in the S1STATES list.

Source DIG2_CDS.CL **Default** \$ord0 **Access Lock** Engineer,
DIG3_CDS.CL build
visible
DIG4_CDS.CL
DIG5_CDS.CL

Type \$3ords, self-defining enumeration of S1STATES

Range \$ord0, \$odr1, \$ord2 (3 or more states only)

PVRAW

Point Type AM Regulatory

Pulses per second

Source M_TZ.CL **Default** 0 **Access Lock** Engineer

Type Real

Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

REFRSHCD()

Point Type AM Custom, Network Point (CLM_nn)

Buffer refresh indicator

Source NETCDS.CL **Default** 0 **Access Lock** Operator

Type Real ARRAY (1..4)

Range 0 – 7

0 = No refresh

1 = Refresh of Schematic CLMSBUF# status data

2 = Refresh of Schematic CLMSBUF# detail status data

3 = Refresh of Schematic CLMPIT# statistical data

4 = Refresh of Schematic CLMGRP# statistical data

5 = Refresh of Schematic CLMPITE# statistical data

6 = Refresh of Schematic CLMGRP# selected statistical data

7 = Refresh of Schematic CLMPITE# selected statistical data

RELNAME

Point Type AM Custom, Network Point (CLM_nn)

Release Control Identification

Source NETCDS.CL **Default** spaces **Access Lock** Program

Type String

RESTART

Point Type AM Regulatory or AM Switch

Restart Type. Note: This is a reserved name that must be represented in CL as RESTART.

Source AM **Default** NONE **Access Lock** Engineer

Type RESTART enumeration

Range NONE, HOT, WARM, COLD, PTACTVN, NOPROC

Continued on next page

2.4 Parameter Name Listing, Continued

**RESTR1,
RESTR2**

Type AM Custom, Network Point (CLM_nn)

Restart CLMI #1, Restart CLMI #2

Source NETCDS.CL **Default** OFF **Access Lock** Operator

Type Logical

Range OFF, ON

S1CUROUT

Point Type AM Switch

Current output calculated from OPy and STx_OPy.

Source DIG2CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG3CDS.CL
DIG4CDS.CL
DIG5CDS.CL

Type \$2ords – \$5ords (Matches the S1NSTATE, number of states)

Range \$ord0 .. \$ord4

S1CURSTS

Point Type AM Switch

Initial or Current operating state of Switch 1

Source AM **Default** \$ord0 **Access Lock** Program

Type \$2ords – \$5ords (Matches the S1NSTATE, number of states)

Range \$ord0 .. \$ord4

S1NSTATE

Point Type AM Switch

Number of valid states for Switch 1

Source AM **Default** 2 **Access Lock** DEB

Type Integer

Range 2 through 5

Continued on next page

2.4 Parameter Name Listing, Continued

S1REQOLD

Point Type AM Switch

Previous requested output

Source DIG2CDS.CL **Default** \$ord0 **Access Lock** Engineer
DIG3CDS.CL
DIG4CDS.CL
DIG5CDS.CL

Type \$2ords – \$5ords (Matches the S1NSTATE, number of states)

Range \$ord0 .. \$ord4

S1REQSTS

Point Type AM Switch

Requested state for Switch 1

Source AM **Default** \$ord0 **Access Lock** Operator
(set by S1ACCLVL)

Type \$2ords – \$5ords (Matches the S1NSTATE, number of states)

Range \$ord0 .. \$ord4

S1STATES(0), S1STATES(1), S1STATES(2), S1STATES(3), S1STATES(4)

Point Type AM Switch

State names for CL Switch states

Source AM **Default** **Access Lock** Engineer-DEB

Type ARRAY (0..4) of String-8

Range Up to eight characters each. Must not be blank.

SALMDSC1, SALMDSC2, SALMDSC3

Point Type AM Switch

Switch alarm 1 to 3 descriptions

Source AM **Default** spaces **Access Lock** Engineer

Type String-8

Continued on next page

2.4 Parameter Name Listing, Continued

SALMFL1,
SALMFL2,
SALMFL3

Point Type AM Switch

Switch alarm flags 1 to 3

Source AM **Default** OFF **Access Lock** View

Type Logical

Range OFF, ON

SCHDCD1,
SCHDCD2,
SCHDCD3,
SCHDCD4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) new schedule code

Source NETCDS1.CL (SCHDCD1) **Default** 0 **Access Lock** Oper
NETCDS2.CL (SCHDCD2)
NETCDS3.CL (SCHDCD3)
NETCDS4.CL (SCHDCD4)

Type Real

Range >= 0.0

SCHDIN1,
SCHDIN2,
SCHDIN3,
SCHDIN4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) schedule address index #

Source NETCDS1.CL (SCHDIN1) **Default** 0 **Access Lock** Operator
NETCDS2.CL (SCHDIN2)
NETCDS3.CL (SCHDIN3)
NETCDS4.CL (SCHDIN4)

Type Real

Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

SP	Point Type AM Regulatory Setpoint in Engineering Units Source AM Default 0 Access Lock Operator Type Real Range SPLOLM to SPHILM
SPPREV	Point Type AM Regulatory Setpoint in Engineering Units on previous scan Source M_AO.CL Default 0 Access Lock Engineer Type Real Range SPLOLM to SPHILM
SPHILM	Point Type AM Regulatory Setpoint high limit in Engineering Units of PV Source AM Default NAN Access Lock Supervisor Type Real Range SPLOLM to SPEXEUHI
SPLOLM	Point Type AM Regulatory Setpoint low limit in Engineering Units of PV Source AM Default NAN Access Lock Supervisor Type Real Range SPEXEULO to SPHILM

Continued on next page

2.4 Parameter Name Listing, Continued

ST0_OP1,
ST0_OP2,
ST0_OP3

Point Type AM Switch

State 0, outputs 1 through 3. Defines the logical state (ON or OFF) that is to be written to output # 1, 2, and 3 when the requested state is state 0.

Source DIG2CDS.CL **Default** OFF Access Lock Engineer
DIG3CDS.CL
DIG4CDS.CL
DIG5CDS.CL

Type Logical

Range OFF, ON

ST1_OP1,
ST1_OP2,
ST1_OP3

Point Type AM Switch

State 1, outputs 1 through 3. Defines the logical state (ON or OFF) that is to be written to output # 1, 2, and 3 when the requested state is state 1.

Source DIG2CDS.CL **Default** OFF Access Lock Engineer
DIG3CDS.CL
DIG4CDS.CL
DIG5CDS.CL

Type Logical

Range OFF, ON

ST2_OP1,
ST2_OP2,
ST2_OP3

Point Type AM Switch

State 2, outputs 1 through 3. Defines the logical state (ON or OFF) that is to be written to output # 1, 2, and 3 when the requested state is state 2.

Source DIG2CDS.CL **Default** OFF Access Lock Engineer
DIG3CDS.CL
DIG4CDS.CL
DIG5CDS.CL

Type Logical

Range OFF, ON

Continued on next page

2.4 Parameter Name Listing, Continued

STACDE()	Point Type AM Custom, Network Point (CLM_nn)
	Buffer station code
	Source NETCDS.CL Default 0 Access Lock Operator
	Type Real ARRAY (1..4)
	Range >= 0.0

START	Point Type AM Switch
	Command disagree timer running
	Source DIG2CDS.CL Default OFF Access Lock Engineer DIG3CDS.CL DIG4CDS.CL DIG5CDS.CL
	Type Logical
	Range OFF, ON

STATIN1, STATIN2, STATIN3, STATIN4	Point Type AM Custom, Network Point (CLM_nn)
	Buffer # (1–4) state address index #
	Source NETCDS1.CL (STATIN1) Default 0 Access Lock Operator NETCDS2.CL (STATIN2) NETCDS3.CL (STATIN3) NETCDS4.CL (STATIN4)
	Type Real
	Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

STATUS

Point Type AM Custom, Builder and Network Points (BUILD_nn and CLM_nn)

Builder Point/Process Status (BUILDCDS)

Status Condition (NETCDS)

Source BUILDCDS.CL **Default** 0 **Access Lock** Engineer
NETCDS.CL

Type Real

Range >= 0.0

STBUFER()

Point Type AM Custom, Network Point (CLM_nn)

Work values

Source NETCDS.CL **Default** 0 **Access Lock** Engineer

Type Real ARRAY (1..20)

Range Any real value

STRNG1(), STRNG2(), STRNG3(), STRNG4()

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) Strings

Source NETCDS1.CL (STRNG1) **Default** spaces **Access Lock** Oper
NETCDS2.CL (STRNG2)
NETCDS3.CL (STRNG3)
NETCDS4.CL (STRNG4)

Type ARRAY (1..150) of String

STRNG1()

Point Type AM Custom, Builder Point (BUILD_nn)

Address Strings

Source CLMDCDS.CL **Default** spaces **Access Lock** Oper

Type ARRAY (1..90) of String

Continued on next page

2.4 Parameter Name Listing, Continued

STSARR()	Point Type AM Custom, Builder Point (BUILD_nn)
	Value Status Array
	Source BUILD_CDS.CL Default "" Access Lock Engineer
	Type ARRAY(1..50) of String
TIMEBASE	Point Type AM Regulatory
	Accumulator time scale factor
	Source M_TZ.CL Default 0 Access Lock Engineer
	Type Real
	Range >= 0.0
TIMER	Point Type AM Switch
	Command disagree timer count down
	Source DIG2_CDS.CL Default 0 Access Lock Engineer DIG3_CDS.CL DIG4_CDS.CL DIG5_CDS.CL
	Type Real
	Range >= 0.0
UPDNIN1, UPDNIN2, UPDNIN3, UPDNIN4	Point Type AM Custom, Network Point (CLM_nn)
	Buffer # (1–4) up down address index #
	Source NET_CDS1.CL (UPDNIN1) Default spaces Access Lock Oper NET_CDS2.CL (UPDNIN2) NET_CDS3.CL (UPDNIN3) NET_CDS4.CL (UPDNIN4)
	Type Real
	Range >= 0.0

Continued on next page

2.4 Parameter Name Listing, Continued

UPRDN1,
UPRDN2,
UPRDN3,
UPRDN4

Point Type AM Custom, Network Point (CLM_nn)

Buffer # (1–4) up down code

Source NETCDS1.CL (UPRDN1) **Default** 0 **Access Lock** Oper
NETCDS2.CL (UPRDN2)
NETCDS3.CL (UPRDN3)
NETCDS4.CL (UPRDN4)

Type Real

Range 0 to 2, 0 for none, 1 for up, 2 for down

VALARR()

Point Type AM Custom, Builder Point (BUILD_nn)

Value Data Array

Source BUILD_CDS.CL **Default** 0 **Access Lock** Engineer

Type ARRAY(1..50) of String

VERROR

Point Type AM Regulatory

Error in validation of entry

Source BUILD_CDS.CL **Default** OFF **Access Lock** Engineer

Type Logical

Range OFF, ON

2.5 Dictionary of File Names

Release file names	This subsection and subsections 2.6 and 2.7 contain complete listings of the files supplied in this release.
BUILD_NN.EB	Initial exception build file for the Builder Interface Point (BUILD_nn)
BUILDER.DS	Builder Main Display source
BUILDER0.DS	Builder Overlay Display source
BUILDER1.DS	Builder Page #1 Overlay Display source
BUILDER2.DS	Builder Page #2 Overlay Display source
BUILDER3.DS	Builder Page #3 Overlay Display source
BUILDER4.DS	Builder Page #4 Overlay Display source
BUILDH01.DS	Builder Help Information Display source
CLM.LO	AM Object Code file for the Platform
CLM.SF	CL Compiler Include file for CLM CL Interface
CLM_NN.EB	Initial exception build file for the Network Interface Point (CLM_nn)
CLMI.S	S Record file for CLMI board resident programs
CLMI.ZI, CLMI.ZT	NDT Template support for the CLMI Interface Board level
CLMI.ZS	Status Definition File Templates for Platform CLMI Board level.
CLMINNN.ZC	Prototype CLMI Load Directive File (use M_CLMI.ZC)

Continued on next page

2.5 Dictionary of File Names, Continued

CLMABORT.DS	Network Display source
CLMDDB.DF	Honeywell Specific Variable Data Area Display source
CLMDETD1.DS – CLMDETD4.DS	Device Detail Status Display source, buffer 1 through buffer 4
CLMGRPD1.DS – CLMGRPD4.DS	Point Interface Table Group (PITG) Display source, buffer 1 through buffer 4
CLMERROR.DS	Invalid Entry to Display (source)
CLMID1.DS	CLM Status Entry Display Subpicture source
CLMMODB0.DO	Display used to support the message point type
CLMPIT1.DS – CLMPIT4	Point Interface Tables (PIT) Status Display source, buffer 1 through buffer 4
CLMPITE1.DS – CLMPITE4.DS	Point Interface Table Entries (PITE) Status Display source, buffer 1 through buffer 4
CLMSBUF1.DS – CLMSBUF4.DS	Network Device Status Display source, buffer 1 through buffer 4
CLMSCHEM.DS	CLM Status Entry Display source
CLMSMT00.DS	BSAP Smartline Transmitter Configuration Display Source
CLMSTAT1.DS	CLM Status Entry Display Source
DIG2.EC, DIG3.EC DIG4.EC DIG5.EC	Unlink/Link CL for 2 through 5 State Digital Composite Points

Continued on next page

2.5 Dictionary of File Names, Continued

DIN2.EC, DIN4.EC	Unlink/Link CL for 2 and 4 State Digital Input Points
GENAPPL.EB	The exception build file for application display points
GENENTER.DS, GENDISPA.DS, GENDISPB.DS, GENDISPC.DS	Display files for the application displays
INSTALL.EC	Platform Installation EC File
M_AI.ZI, M_AI.ZT	Templates supporting analog input image points using the Modbus protocol
M_AO.ZI, M_AO.ZT	Templates supporting analog output image points using the Modbus protocol
M_CLMI.ZC	Prototype CLMI Load Directive File for Modbus
M_DEVICE.ZI, M_DEVICE.ZT	NDT Templates support for the Modbus Devices
M_DEVICE.ZS	Status Definition File Templates for Modbus
M_DIN2.ZI, M_DIN2.ZT	Templates supporting two state digital image points using Modbus with a single input
M_DIN4.ZI, M_DIN4.ZT	Templates supporting four state digital image points using Modbus with two inputs
M_D2.ZI, M_D2.ZT	AM digital composite point (2 states) using Modbus Digital I/O
M_D4.ZI, M_D4.ZT	AM digital composite point (4 states) using Modbus Digital I/O
M_FLAG.ZI, M_FLAG.ZT	Templates to implement a Modbus logical signal as an AM Flag Point

Continued on next page

2.5 Dictionary of File Names, Continued

M_INST.EC	Network Interface Layer and Protocol Installation EC for Modbus
M_MSG.ZI, M_MSG.ZT	Templates to implement a message interface using Modbus signals and an AM Custom Point
M_NUMER.ZI, M_NUMER.ZT	Templates to implement a Modbus signal in an AM Numeric Point
M_PORT.ZI, M_PORT.ZT	NDT Template support for the Port level using Modbus
M_PORT.ZS	Status Definition File Templates for Modbus Port level
M_TZ.ZI, M_TZ.ZT	Templates to implement a Modbus accumulator in an AM Regulatory Point
MODB.S	S record file for Modbus Protocol
PITG.ZI, PITG.ZT	Template support for Point Interface Table Group
PIT.ZI, PIT.ZT	Template support for Point Interface Table

2.6 Summary of CL Data Segments

BLD2CDS.CL, BUILDCDS.CL	CDS Blocks for CLM Builder Status Point (BUILD_nn). Provides support for the Builder process.
CLM_DMD.CL	CDS for CLM Platform User CL Interface. Includes parameters CLM_DMD, CLM_OUT, CLM_PPS, CLM_PIT, CLM_PORT and CLM_DEV Parameters. Parameters are NOT BLD_VISIBLE, require Engineer ACCESS level, and appear on the CDS page. After running the builder, you must save this data in the AM checkpoint files.
CLM_DMD2.CL	CDS for CLM Platform User CL Interface. Includes parameters CLM_DMD, CLM_OUT, CLM_PPS, CLM_PIT, CLM_PORT and CLM_DEV Parameters. Parameters are BLD_VISIBLE, require Engineer ACCESS level, and appear on the CDS page. After running the builder, you must save this data in the AM checkpoint files.
CLMDCDS.CL	CDS for CLMI memory read/write function. To be fully implemented in Platform Release 230.
DIG2CDS.CL, DIG3CDS.CL, DIG4CDS.CL, DIG5CDS.CL	Image point parameters for Generic 2- to 5-state composite digital I/O points. Parameters are NOT BLD_VISIBLE and require Engineer Access level. Exceptions on build visibility and access level are noted in the foregoing dictionary. In switch points, all CDS parameters appear on the CDS page.
DIN2CDS.CL, DIN4CDS.CL	Image point parameters for Generic 2- and 4-state composite digital input points. Parameters are NOT BLD_VISIBLE and require Engineer Access level. Exceptions on build visibility and access level are noted in the foregoing dictionary. In switch points, all CDS parameters appear on the CDS page.

Continued on next page

2.6 Summary of CL Data Segments, Continued

GENRLCDS.CL

This is the Custom Data Segment (CDS) for CLM Application processing. This will support all CLM Applications functionality for picture processes. The application developer can use this CDS to pass values to and from the user. Parameters are not build visible and require Engineer Access Level. In AM Custom points, all parameters appear on the CDS page. It is made up of arrays and will be attached to a special Port LCN point for the application executing on that port.

The point ID will be in the format of "CLMnnPp–" where:

- nn = the CLM node number,
- p = the Application Port number, and
- the – = A, B, and C.

The process requires three separate Custom Points to hold all of the values involved. The CDS is the same for each point, but some values will only be used from the point with the "A" code identifier.

M_AI.CL

CDS Image point parameters for the Modbus analog input image point. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the PV algorithm page.

M_AO.CL

CDS Image point parameters for the Modbus analog output image point. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the Control algorithm page.

M_MSG.CL

CDS Image point parameters for the Modbus message interface image point. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the CDS page.

M_TZ.CL

CDS Image point parameters for the Modbus accumulator image point. Parameters are NOT BLD_VISIBLE, require Engineer Access level, and appear on the PV algorithm page.

Continued on next page

2.6 Summary of CL Data Segments, Continued

NETCDS.CL

The first CDS Block for CLM Network Status Point. This is the Custom Data Segment (CDS) for CLM processing. This will support all CLM Network functionality for the Network level. Other CDSs (see the next topic) are required for complete functionality.

**NETCDS1.CL,
NETCDS2.CL,
NETCDS3.CL,
NETCDS4.CL**

CLM Network Status Point CDS for buffer 1 through 4.

2.7 Summary of CL Programs

DIG2CL.AO

Compiled CL routines for 2-State Digital Composite Switch Points.

Common CL Procedure Blocks (object)

Generic Block at the general insertion point (10) named DIG2CL.

ERR2 validates parameters ALMOPT, NODOPTS, and NODINPTS for BADVAL and correct range. Corrects FBTIME if BADVAL. Validates S1NSTATE=2. Invalid configuration is flagged by SALMFL2. Alarms and outputs are blocked by invalid configuration. Parameters with BADVAL are corrected to -1.

MAP2 determines S1CURSTS from inputs and mapping parameters.

ALARM2 off-normal alarming. Uses SALMFL1.

OUTPUT2 Uses output mapping parameters. If NODINPTS = 0, determines S1CURSTS from read back values of OP1 and OP2. Determines OP1 and OP2 from S1REQSTS. Command disagree alarming.

CDALARM2 Command disagree alarm timer.

DIG3CL.AO

Compiled CL routines for 3-State Digital Composite Switch Points.

Included is one Generic Block at the general insertion point (10) named DIG3CL.

ERR3 validates parameters ALMOPT, NODOPTS, and NODINPTS for BADVAL and correct range. Corrects FBTIME if BADVAL. Validates S1NSTATE=3. Invalid configuration is flagged by SALMFL2. Alarms and outputs are blocked by invalid configuration. Parameters with BADVAL are corrected to -1.

MAP3 determines S1CURSTS from inputs and mapping parameters.

ALARM3 off-normal alarming. Uses SALMFL1.

OUTPUT3 Uses output mapping parameters. If NODINPTS = 0, determines S1CURSTS from read back values of OP1, OP2, and OP3. Determines OP1, OP2, and OP3 from S1REQSTS.

Command disagree alarming.

CDALARM3 Command disagree alarm timer.

Continued on next page

2.7 Summary of CL Programs, Continued

DIG4CL.AO

Compiled CL routines for 4-State Digital Composite Switch Points. Included is one Generic Block at the general insertion point (10) named DIG4CL.

ERR4 validates parameters ALMOPT, NODOPTS, and NODINPTS for BADVAL and correct range. Corrects FBTIME if BADVAL. Validates S1NSTATE=4. Invalid configuration is flagged by SALMFL2. Invalid command is flagged by SALMFL3. Alarms and outputs are blocked by invalid configuration. Outputs are blocked by the invalid command alarm. Parameters with BADVAL are corrected to -1.

MAP4 determines S1CURSTS from inputs and mapping parameters.

ALARM4 off-normal alarming. Uses SALMFL1.

OUTPUT4 Uses output mapping parameters. If NODINPTS = 0, determines S1CURSTS from read back values of OP1, OP2, and OP3. Determines OP1, OP2, and OP3 from S1REQSTS.

Command disagree alarming.

CDALARM4 Command disagree alarm timer.

DIG5CL.AO

Compiled CL routines for 5-State Digital Composite Switch Points. Included is one Generic Block at the general insertion point (10) named DIG5CL.

ERR5 validates parameters ALMOPT, NODOPTS, and NODINPTS for BADVAL and correct range. Corrects FBTIME if BADVAL. Validates S1NSTATE=5. Invalid configuration is flagged by SALMFL2. Invalid command is flagged by SALMFL3. Alarms and outputs are blocked by invalid configuration. Outputs are blocked by the invalid command alarm. Parameters with BADVAL are corrected to -1.

MAP5 determines S1CURSTS from inputs and mapping parameters.

ALARM5 off-normal alarming. Uses SALMFL1.

OUTPUT5 Uses output mapping parameters. If NODINPTS = 0, determines S1CURSTS from read back values of OP1, OP2, and OP3. Determines OP1, OP2, and OP3 from S1REQSTS.

Command disagree alarming.

CDALARM5 Command disagree alarm timer.

Continued on next page

2.7 Summary of CL Programs, Continued

DIN2CL.AO

Compiled CL routines for 2-State Digital Input Switch Points.

Common CL Procedure Blocks (object)

Generic Block at the general insertion point (10) named DIN2CL.

ERR2 validates parameter ALMOPT for BADVAL and correct range.

Validates S1NSTATE=2. Invalid configuration is flagged by SALMFL2. Alarms are blocked by invalid configuration.

MAP2 determines S1CURSTS from inputs and mapping parameters.

ALARM2 off-normal alarming. Uses SALMFL1.

DIN4CL.AO

Compiled CL routines for 4-State Digital Input Switch Points. Included is one Generic Block at the general insertion point: (10) named DIN4CL.

ERR4 validates parameter ALMOPT for BADVAL and correct range.

Validates S1NSTATE=4. Invalid configuration is flagged by SALMFL2. Alarms are blocked by invalid configuration.

MAP4 determines S1CURSTS from inputs and mapping parameters.

ALARM4 off-normal alarming. Uses SALMFL1.

M_AOCL.AO

Compiled CL routines for Modbus analog output Points. Included is one Generic Block at the general insertion point (10) named M_AOCL.

The block is not executed when the point is in the MAN mode.

The block handles initialization and does outputs following setpoint changes.

Continued on next page

2.7 Summary of CL Programs, Continued

M_MSGCL.AO

Compiled CL routines for Modbus message points. Included is one Generic Block at the general insertion point (10) named M_MSGCL.

On initialize, the two input strings will be set to null.

The CL block moves a non-null input string from INPUT to LAST_IN and adds it to the push down list IOARRAY. The date/time is added to the push down list IOTIME. An array (OUTFLAG) discriminates between input and output.

If the user has set the parameter JOURNAL to ON, then a message will be sent to the operator.

The user should use a display (CLMMODB0) to enter a value for OUTPUT, to view date, and other items not visible on the detail display.

M_TZCL.AO

Compiled CL routines for accumulator or totalizer points. Included is one Generic Block at the general insertion point (10) named M_TZCL.

The block is not executed when the point is in the MAN mode.

The block validates the user's configuration of C1, C2, and TIMEBASE.

Pulses per second are computed and saved in PVRAW.

Engineering units per time are computed and saved in PCALC.

READER COMMENTS

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Writer: **Ken Arnold**

COMMENTS: _____

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